

$$= \frac{\sum_{s=1}^t \lambda_s P_s^-(x) D_s(y|x) + P_s^+(x) S_s^+(y|x)}{\sum_{s=1}^t \lambda_s P_s^-(x) + P_s^+(x)}$$

and therefore

$$\begin{aligned} \bar{q}_t(x) &= \sum_y \mathcal{D}_t(y|x) q(x, y) \\ &= \frac{\sum_y \sum_{s=1}^t \lambda_s P_s^-(x) D_s(y|x) q(x, y) + P_s^+(x) S_s^+(y|x) q(x, y)}{\sum_{s=1}^t \lambda_s P_s^-(x) + P_s^+(x)} \\ &= \frac{\sum_{s=1}^t \lambda_s P_s^-(x) q_s(x) + P_s^+(x) q_s^+(x)}{\sum_{s=1}^t \lambda_s P_s^-(x) + P_s^+(x)} \quad \square \end{aligned}$$

Lemma 7. For all $t \geq 1$ and $x \in P$ we have $\bar{q}_t(x) = 0$ if and only if $\bar{q}_{t-1}(x) = 0$ and $q_t(x) = 0$.

Proof. Suppose $\bar{q}_t(x) = 0$. By Lemma 6 this implies $\lambda_s P_s^-(x) q_s(x) + P_s^+(x) q_s^+(x) = 0$ for $s \in \{1, \dots, t\}$, and therefore $\bar{q}_{t-1}(x) = 0$. This implies that x cannot be correctly labeled in S_t , and therefore $P_t^-(x) = 1$. Since $\alpha > 0$ we have $\lambda_t > 0$. And since $\lambda_t P_t^-(x) q_t(x) = 0$ we must have $q_t(x) = 0$.

Now suppose $\bar{q}_{t-1}(x) = 0$ and $q_t(x) = 0$. Since $\bar{q}_{t-1}(x) = 0$ then again by Lemma 6 we have $\lambda_s P_s^-(x) q_s(x) + P_s^+(x) q_s^+(x) = 0$ for $s \in \{1, \dots, t-1\}$. The fact that $\bar{q}_{t-1}(x) = 0$ also implies that x cannot be correctly labeled in S_t , and therefore $P_t^+(x) = 0$. And since $q_t(x) = 0$ we have $\lambda_t P_t^-(x) q_t(x) + P_t^+(x) q_t^+(x) = 0$, which implies $\bar{q}_t(x) = 0$. $\square$

Lemma 8. Let $t \geq 1$ and $x \in P$. If $x \in P_t^-$ then $q_t(x) \in \{0, 1\}$. If $x \in P_t^+$ then $q_t^+(x) = 1$.

Proof. Note that D_t contains each prompt only once (by Definition 3 of `labeler $_\beta$`), and S_t^+ contains only correctly labeled prompts (by Definition 2 of `filter`). The lemma follows from the definitions of P_t^- , P_t^+ , $q_t(x)$ and $q_t^+(x)$. $\square$

Lemma 9. If $a, b, c, d \geq 0$ satisfy $a \leq b$, $c \geq d$ and $b > 0$ then

$$\frac{a+c}{b+c} \geq \frac{a+d}{b+d}.$$

Proof. If $c = d$ then clearly the lemma holds with equality. Otherwise if $c > d$ then

$$\begin{aligned} \frac{a+c}{b+c} &\geq \frac{a+d}{b+d} \\ \Leftrightarrow (a+c)(b+d) &\geq (a+d)(b+c) && b > 0 \\ \Leftrightarrow ab + bc + ad + cd &\geq ab + bd + ac + cd \\ \Leftrightarrow bc + ad &\geq bd + ac \\ \Leftrightarrow b(c-d) &\geq a(c-d) \\ \Leftrightarrow b &\geq a && c > d \quad \square \end{aligned}$$

Our analysis relies on conditioning on the fact that once the quality of a particular prompt, x , is high enough, it is always selected by `filter` and is never sent to `labeler $_\beta$` . Formally, fix the number of iterations, T , the set of prompts, P , and the quality of the weak data, β . We define event E , as follows:

Event $E \equiv$ For all $t \in [T]$ and $x \in P$ if $\bar{q}_{t-1}(x) \geq \beta$ then $x \notin P_t^-$,

and to simplify notation, we drop the dependence of E on T , P and β .

Lemma 10. *If the repeat parameter $k \geq \frac{2 \log T + \log |P|}{\beta \gamma}$ then event E occurs with probability at least $1 - \frac{1}{T}$.*

Proof. By the definition of **generate** (Definition 4), each $x \in P$ is labeled k times by g_{t-1} in iteration t , with each label drawn independently from distribution $g_{t-1}(x)$. Thus we know that if $\bar{q}_{t-1}(x) \geq \beta$ then $x \in P_t^-$ with probability at most $(1 - \beta \gamma)^k$. This is because the probability that a synthetic label does not prevent a prompt from being added to P_t^- is at most $1 - \beta + \beta(1 - \gamma) = 1 - \beta \gamma$, which is the probability that the label is low-quality *or* that it is high-quality but is not recognized by **filter** $_\gamma$ (Definition 2). Therefore

$$\begin{aligned}
\Pr[\neg E] &= \Pr[\exists t \in [T] \text{ and } x \in P \text{ such that } \bar{q}_{t-1}(x) \geq \beta \text{ and } x \in P_t^-] \\
&\leq \sum_{t=1}^T \sum_{x \in P} \Pr[\bar{q}_{t-1}(x) \geq \beta \text{ and } x \in P_t^-] \\
&\leq T|P|(1 - \beta \gamma)^k \\
&\leq T|P|\exp(-\beta \gamma k) \\
&\leq T|P|\exp(-2 \log T - \log |P|) && \text{Assumption about } k \\
&= T|P|\frac{1}{T^2} \frac{1}{|P|} \\
&= \frac{1}{T}
\end{aligned}$$

□

The next result is our key lemma. It says that if event E occurs then (a) P_t^- contains all and only the prompts that must have been incorrectly labeled by the previous iteration's LLM, (b) the size of P_t^- shrinks exponentially over time, (c) once a prompt is outside P_t^- it remains that way, and (d) prompts outside of P_t^- are correctly labeled by the previous iteration's LLM with a probability that is bounded above zero.

Lemma 11. *Fix T . Let $1 \leq t \leq T$ and $x \in P$. If event E occurs then all of the following hold:*

- (a) $x \in P_t^-$ if and only if $\bar{q}_{t-1}(x) = 0$.
- (b) $|P_r^-| \leq (1 - \beta)^{r-s} |P_s^-|$ for all $r, s \in [t]$ such that $r \geq s$.
- (c) There exists $r \in [t]$ such that $x \in P_s^-$ for all $s \in [r]$ and $x \notin P_s^-$ for all $s \in [t] \setminus [r]$.
- (d) Let $r \in [t]$ satisfy the conditions of part (c). If $r < t$ then

$$\bar{q}_t(x) \geq \frac{\alpha + t - r}{\frac{\alpha(1-(1-\beta)^r)}{\beta} + t - r} \geq \beta.$$

Proof. The proof will proceed by induction. We begin by proving the base case, $t = 1$. To prove part (a), note that by assumption we have $\bar{q}_0(x) = \mathbb{E}_{y \sim g_0(x)}[q(x, y)] = 0$, so we only need to show that $x \in P_1^-$. Since $\mathbb{E}_{y \sim g_0(x)}[q(x, y)] = 0$, we know that x cannot be correctly labeled in S_1 , which implies

$x \in P_1^-$. Part (b) follows immediately from the observation that when $t = 1$ we have $r = s = 1$. Part (c) holds immediately by letting $r = 1$, since in this case $[t] \setminus [r]$ is empty, and we have already shown $x \in P_1^-$ in part (a). Part (d) holds vacuously because $r < t$ must be false when $t = 1$.

Now assume for induction that the lemma holds for a fixed $t \geq 1$. We will prove the lemma for the case $t + 1$. To prove part (a), first assume $\bar{q}_t(x) = 0$, which is the premise of the ‘if’ direction. By Definition 1 we have

$$\mathbb{E}_{y \sim g_t(x)}[q(x, y)] = \sum_y \mathcal{D}_t(x, y)q(x, y) = \bar{q}_t(x) = 0$$

which implies that x cannot be correctly labeled in S_{t+1} , and therefore $x \in P_{t+1}^-$. Now assume $x \in P_{t+1}^-$, which is the premise of the ‘only if’ direction. To force a contradiction, assume that $\bar{q}_t(x) > 0$. By part (d) of the inductive hypothesis, this implies $\bar{q}_t(x) \geq \beta$. Since event E occurred, we have that $x \notin P_{t+1}^-$, which is a contradiction. This completes the proof of part (a).

To prove part (b), choose any $r, s \in [t + 1]$ such that $r \geq s$. If $r = s$, part (b) follows immediately. If $r < t + 1$ and $s < t + 1$ then part (b) follows from the inductive hypothesis. Henceforth assume $s < r = t + 1$. Let

$$D_t^+ = \{(x, y) \in D_t : q(x, y) = 1\}$$

be the subset of D_t that is correctly labeled. We have

$$\begin{aligned} (1 - \beta)|P_t^-| &= (1 - \beta)|D_t| \\ &\geq |D_t| - |D_t^+| && \text{Definition 3 of } \texttt{labeler}_\beta \\ &= \sum_{x,y} D_t(x, y) - \sum_{x,y} D_t(x, y)q(x, y) \\ &= \sum_{x,y} D_t(x, y)(1 - q(x, y)) \\ &= \sum_{x,y} P_t^-(x)D_t(y|x)(1 - q(x, y)) \\ &= \sum_x P_t^-(x)(1 - q_t(x)) \\ &= \sum_x P(x)\mathbf{1}\{\bar{q}_{t-1}(x) = 0\}(1 - q_t(x)) && \text{Inductive hypothesis, part (a)} \\ &= \sum_x P(x)\mathbf{1}\{\bar{q}_{t-1}(x) = 0\}\mathbf{1}\{q_t(x) = 0\} && \text{Lemma 8} \\ &= \sum_x P(x)\mathbf{1}\{\bar{q}_t(x) = 0\} && \text{Lemma 7} \\ &= \sum_x P_{t+1}^-(x) && \text{Part (a)} \\ &= |P_{t+1}^-| \end{aligned}$$

and therefore

$$|P_{t+1}^-| \leq (1 - \beta)|P_t^-| \leq (1 - \beta)(1 - \beta)^{t-s}|P_s^-| = (1 - \beta)^{t+1-s}|P_s^-| = (1 - \beta)^{r-s}|P_s^-|$$

where the second inequality follows from the inductive hypothesis. This completes the proof of part (b).